

Haram Naseeb

Backend & IoT Systems Engineer

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Summary

Computer Science student with hands-on experience in backend systems, IoT development, and AI simulations. Proficient in C#, Python, JavaScript, and SQL with a strong foundation in OOP, database architecture, and full-stack project delivery. Seeking backend or IoT engineering internships where I can contribute to real-world systems while deepening expertise in distributed and embedded software.

Projects

Smart IoT Notice Board System

Spring 2025

ESP32 MQTT Node.js MySQL WiFi

- **Role: Firmware & Communication Developer**
- Worked on ESP32 firmware integration to handle WiFi connectivity and real-time MQTT-based message reception from the broker.
- Implemented real-time data exchange between the hardware device and the web dashboard using the publisher-subscriber messaging model.
- Contributed to hardware-software integration and participated in system debugging across the embedded and network communication layers.

AI Simulation Engine

Spring 2025

Python C# OOP SOLID

- **Role: Problem Simulation & Algorithm Implementation Developer**
- Implemented AI problem simulation environments for 8-Queens, Vacuum Agent, and Wumpus World, including state representation and transition logic for each.
- Integrated core algorithms (backtracking, search logic) into the simulation flow to drive agent decision-making within each environment.
- Built a step-by-step execution system that visualises the AI decision process at each state transition, making algorithm behaviour directly observable.
- Ensured a modular OOP-based architecture so each problem environment is independently reusable and extensible across the codebase.

Database Learning Platform

Spring 2025

React PostgreSQL MongoDB Neo4j Node.js

- **Role: Frontend & Learning System Developer**
- Developed the full frontend interface in React, building interactive DBMS learning modules with a structured, responsive UI flow.
- Built a quiz engine with scoring, automated evaluation, and per-user progress tracking across multiple database learning modules.
- Implemented a certificate generation module that issues completion certificates upon passing course assessments.
- Integrated multi-database learning modules (PostgreSQL, MongoDB, Neo4j) into a unified frontend navigation and content structure.

StegoSec Security Simulation Platform

Spring 2025

React JWT IndexedDB JavaScript AES-256

- **Role: Backend, Authentication & Security Module Developer**
- Implemented the backend authentication system using JWT with role-based access control, managing distinct permissions for admin and student users.
- Designed and managed the database structure for users, audit logs, and encrypted messages stored persistently via IndexedDB.

- Built attack simulation modules covering AES encryption/decryption walkthroughs and network interception analysis tools.
- Developed an admin-level logging and activity tracking system for real-time security event monitoring across user sessions.

Airline Management System

Spring 2024

C# .NET WinForms MySQL ADO.NET

- **Role: Full-Stack Developer (team of 2)** — handled both frontend and backend development across the full system lifecycle.
- Designed and implemented a strict 3-layer architecture: UI (WinForms), Business Logic Layer, and Data Access Layer using ADO.NET.
- Developed flight, booking, and passenger management modules with full MySQL integration and a normalised relational database schema.
- Owned the complete system workflow including database design, application logic, and integration between all architectural layers.

DSA Visualization System

Fall 2025

C# .NET OOP

- **Role: Frontend, Authentication & Analytics Module Developer**
- Developed frontend modules including the user authentication system and personal profile management interface.
- Implemented a leaderboard and ranking system driven by data structure-based logic to surface user performance standings.
- Built interactive analytics sections for visualising user progress and performance metrics across completed exercises.
- Contributed UI components for algorithm demonstration and interactive learning features within the broader 4-person team codebase.

Technical Skills

Languages	C#, Python, JavaScript, SQL, C
Backend / DB	Node.js, Express, MySQL, PostgreSQL, Neo4j, MongoDB, IndexedDB, JWT
IoT / Embedded	ESP32, Arduino, MQTT (HiveMQ, EMQX), WiFi communication
Frontend	React, HTML5, CSS3, responsive UI design
Concepts	OOP, SOLID principles, system design, DSA, REST APIs, layered architecture
Tools	Git, GitHub, VS Code, Visual Studio, Arduino IDE

Education

University of Engineering & Technology, Lahore
BS Computer Science

Expected 2028
CGPA: 3.86 / 4.00 — *Dean's List*

Achievements & Involvement

- **HackerRank Certifications:** SQL (Basic), SQL (Intermediate), SQL (Advanced)
- **Hackathons:** Codex 3.0, ACM Coding Competition
- **Organizations:** ACM Student Chapter, Google Developer Group (GDG) UET Lahore
- **ITEC 25:** Non-Technical Management & Coordination Lead
- **Events Coordinator:** UET Lahore student events